

# AI 507: Artificial Intelligence and Society



## #9 The Internet of Things

Associate Prof. Dr. Lena Frischlich

# Flashlight

# Flashlight

## Chatbots

Laswell formula for communication

Turing test

Eliza

GPTs

## Malicious social bots

Coordinated inauthentic behavior

Superspreaders

From bot or not to behavioral patterns

Spiral of Silence

## AI-chatbots for good

Habermasian machine

Counterspeech

AI for news dissemination

## Intimate relationships with chatbots

Self-disclosure and relationship formation

Prevalence of chatbot relations

Favourability of chatbot relations

Intimacy by design & over disclosure by chatbots

**Any questions on that?**



# Then let's get started!

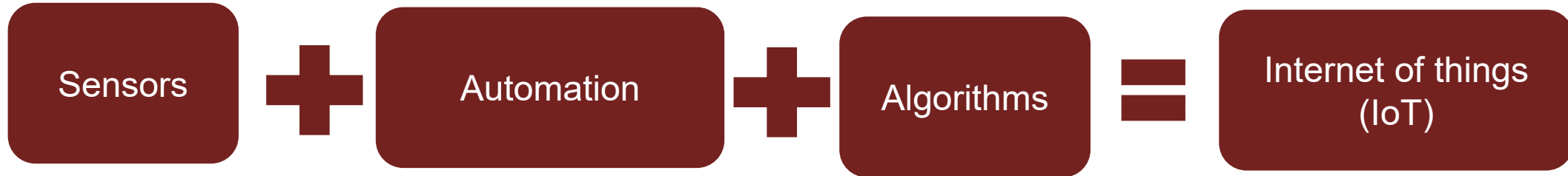
After today's lecture, you will hopefully...

- ☐ ...have a better understanding of smartification
- ☐ ... are aware of security challenges due to the Internet of Things
- ☐ ...have discussed how people handle privacy in digital environments –  
and how the IOT impairs this response

# Smartification

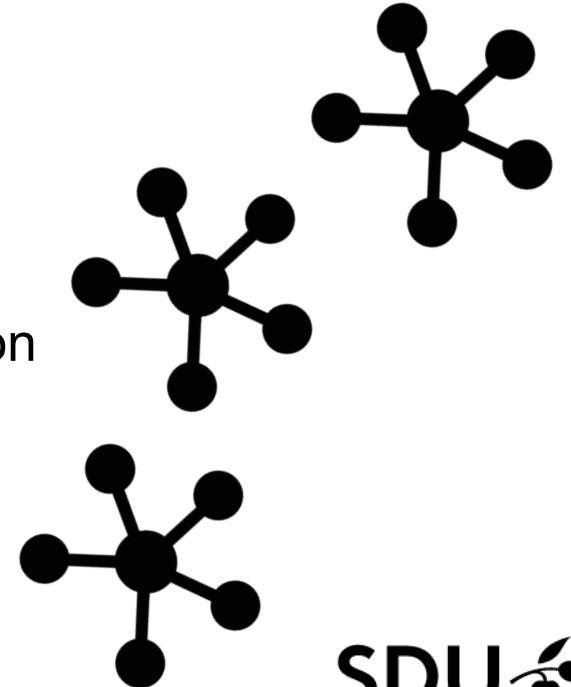
# Smartification

(Van Deursen & van Dijk, 2019)



The *Internet of Things* describes everyday objects that are connected to the Internet and that are...

- equipped with sensors, storage facilities and processing capacities that enable an understanding of the environment,
- equipped with identification and networking capabilities that allow information to be communicated
- capable of autonomous decision-making



# The idea is already a few years old

(cf Rose et al., 2015)

- As early as 1968, the Greek-American inventor Theodore Paraskevakos was working on the transmission of electronic signals over the telephone line
- A technique that became the basis of the ID that now tells you who is calling you
  - In 1999, the British tech pioneer Kevin Ashton coined the term
  - “Illustration of
  - possibilities” for the use of radio-frequency identification (RFID) tags which can be used to digitally monitor the number of goods (e.g., of packages leaving the factory)



Theodore G. Paraskevakos  
Via Wikipedia



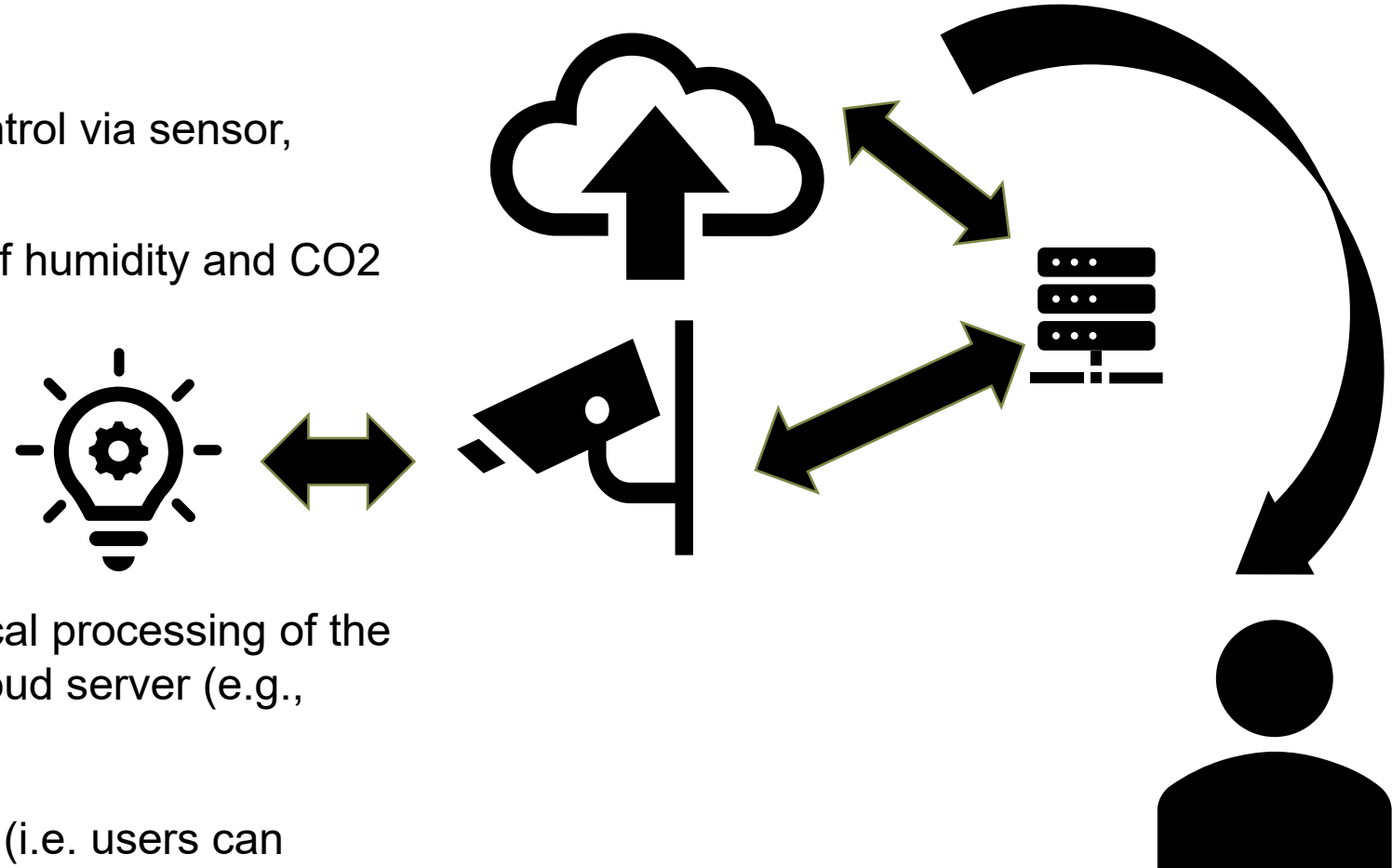
Kevin Ashton via Wikipedia

# Generally: four central communication modes

(Rose et al., 2015)

Device-to-device (e.g., lighting control via sensor, refrigerator to supermarket)

Device-to-cloud (e.g., integration of humidity and CO2 sensors in a digital cloud)



Device to gateway to cloud, i.e. local processing of the data and then forwarding to the cloud server (e.g., smartphone apps, fitness trackers)

Extension: Back-end data sharing (i.e. users can download, aggregate and analyse data from different clouds)

# Today

(Hansen et al., 2024; Foodnationdenmark, *n.d.*)

Internet of Things (IoT)

Wearables

Which smartified  
objects do you  
know/ use?

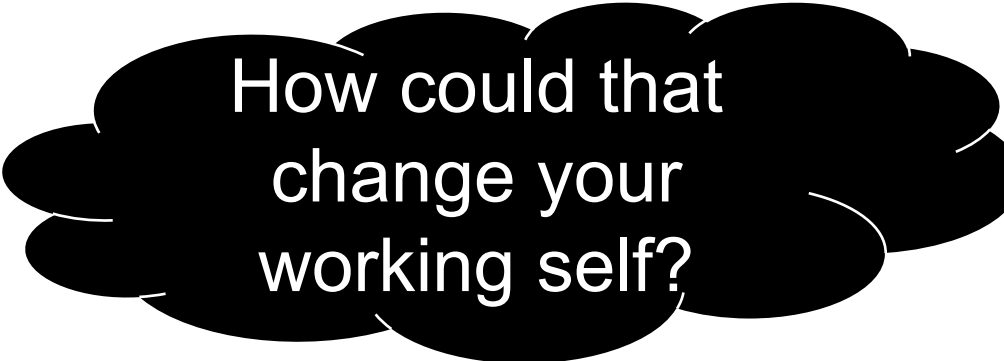
# Self-tracking as a laboratory of the self

(Kristensen & Ruckenstein, 2018)

- Recall #4 Social Cognition: Our self-concept is situationally encoded – we do understand ourselves through our response to (social) situations (→ “working self-concept”)
- Wearables now allow new answers to old question “who am I”

Ethnographic work in the Danish self-tracking community

- Focus on fewer "fixed" members
- Self-Tracking as
  - Objectification (me as my data)
  - subjectivation (in which situation do I react how?)
- Both: New possibilities – but also technological limits to what can be tracked
- Spectrum between self-discovery and self-improvement



How could that  
change your  
working self?

# Today

(Hansen et al., 2024; Foodnationdenmark, *n.d.*)

Internet of Things (IoT)

Wearables

Homes

Cities

Transport

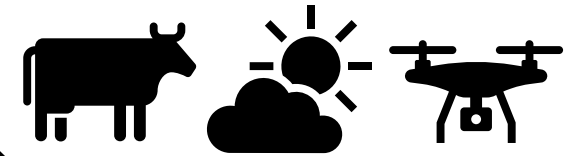
Farming

Industry



- 1 in 2 Danes had at least one smart technology at home in 2020/2021
- Mostly voice-control or music system (30%)
- Followed by security (24%)

Which smartified objects do you know/ use?



71% of Danish farmers use RTK-GPTS

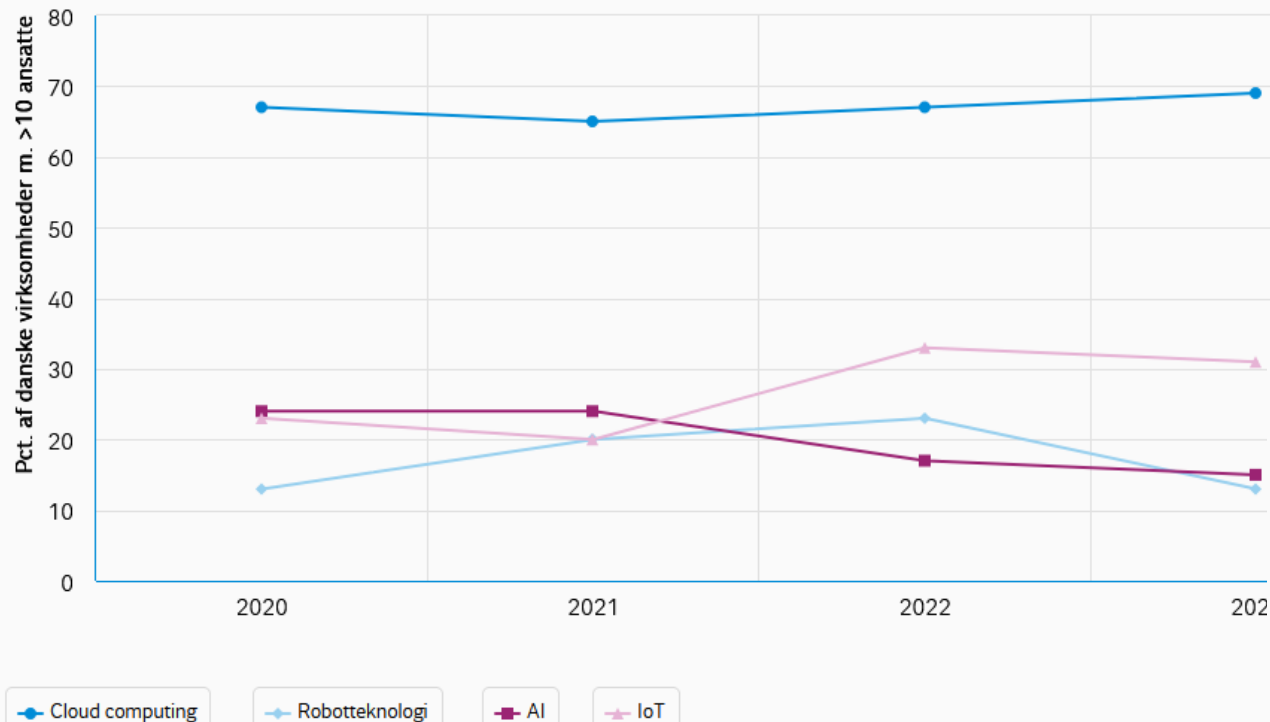
33% use drones and satellites

# In Denmark

(Danskindustri, n.d.)

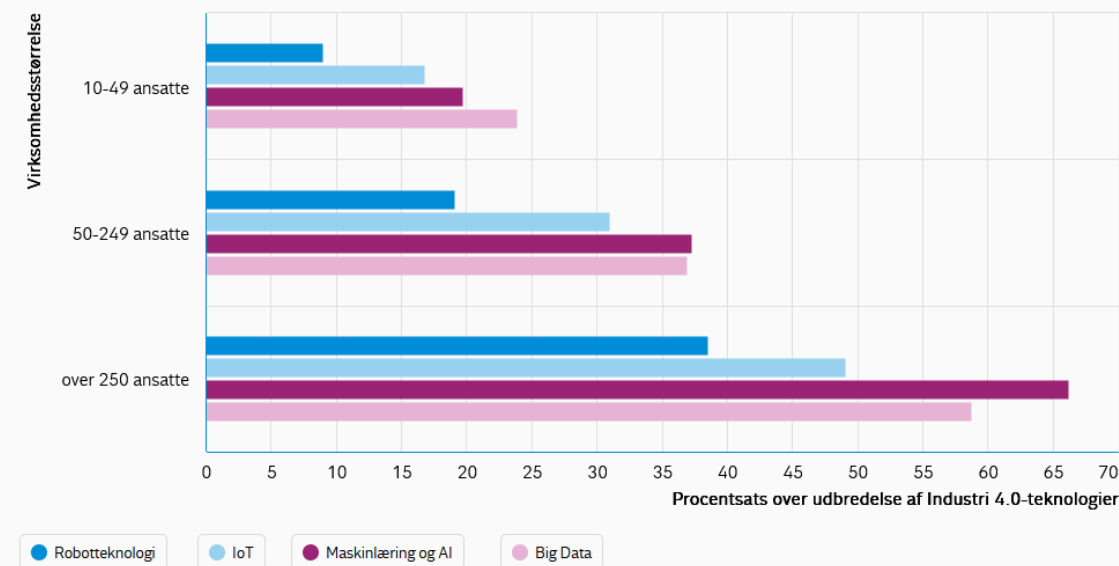
## Danske virksomheders brug af Industri 4.0-teknologier

Data rekvireret efter egen tilvirkning på baggrund af tal fra dst.dk.



## Udbredelse af Industri 4.0-teknologier

i danske virksomheder fordelt på mængden af ansatte. Data rekvireret på baggrund af Desk research fra IRIS Group.



# Motivations and barriers to smartification

(Stentoft et al., 2017)

- Survey with 570 Danish Companies
- Overall good knowledge about smart technologies
- But: Knowledge bigger in larger companies and its mostly bigger companies who implement smart technologies

## Motivators

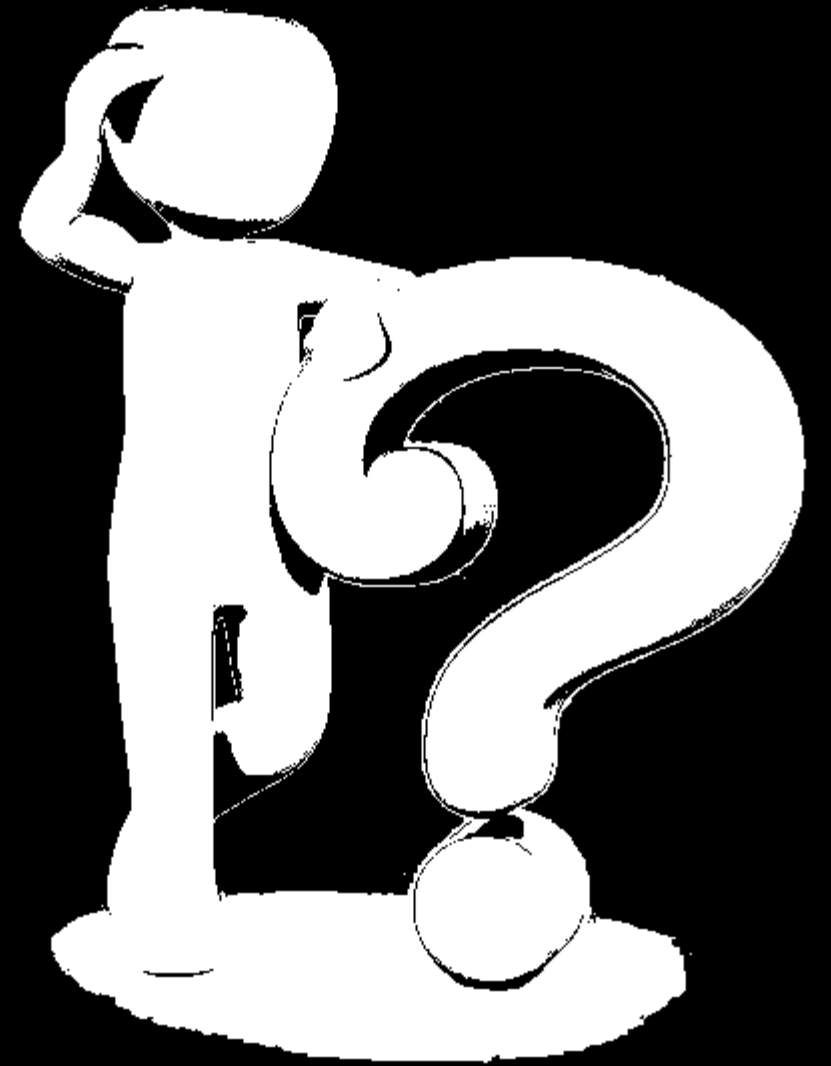
- (1) Cost reduction
- (2) Improvement of time-to-market
- (3) Legal/legislative conditions

## Barriers

- (1) Lack of knowledge
- (2) Focus on operation (development is more smartified)
- (3) Lack of understanding of the strategic importance

→ #5 Technology Acceptance Model  
(perceived usefulness & perceives ease of use)

**Questions: Write me on  
itslearning**



# The Internet of Things and Security

# Smart devices promise security

(Jha, 2022; Paiva et al., 2023; Paulsen, 2022)

74%

of users cite  
safety as a motive  
~ 31% of  
Germans  
(Paulsen, 2022)

567

Millionen \$  
Sales are expected  
for baby monitors in  
2022 (Jha, 2022)

- Growing interest in how they could support victims of sexual abuse through
  - Automated defenses (shocking perpetrators)
  - Releasing alarms
  - Collecting evidence (e.g., physiological data, geo-location data)
  - BUT: Very individual physiological reactions and behavioural responses to abuse  
(→ Gisele Pelicot was at home and “sleeping”)

# But: also smarter attacks

(Crawford, 2024; Kumari & Jain, 2023; Moore, 2022)



- DDOS attacks via social bots herded via small smartified household items
- E.g., Mirai Botnet with ~ 3 million IoT device bots → has brought down e.g., Netflix, CNN, Twitter, Reddit
- Apple AirTags are small devices that you can use to find lost items
- Not connected to the Internet but emit a Bluetooth signal that other Apple devices can detect
- Has facilitated stalking and harassment from abusers
- In one case in the US used by a women to detect her boyfriend and run over him multiple times



# Why is the IoT so special?

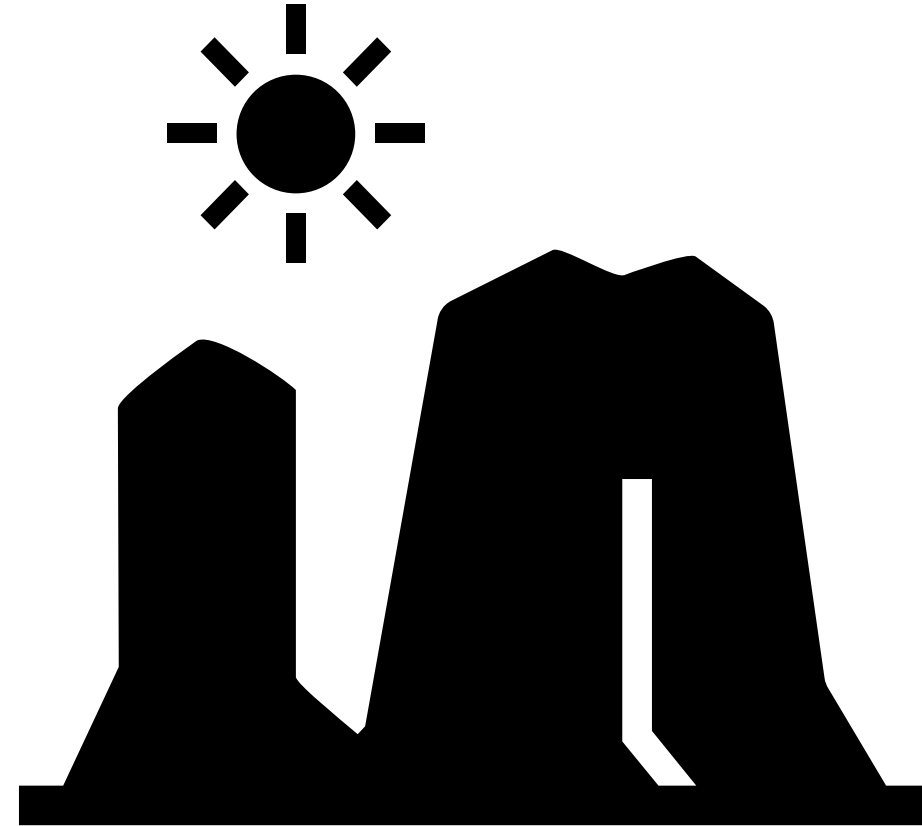
(Rose et al., 2015)

- Mass production of sensors
- Homogeneity: Mass production of very similar devices makes a single vulnerability exploitable on a large scale
- Longevity: Long-term updates, etc. must be granted (e.g., renewal of your refrigerator)
- Lack of transparency: For users, it often remains completely unclear which data is processed and how or to whom
- Embedding: Urban/agricultural sensors may not be directly visible, observing
- No possibility of objection
- Risks are often unequally distributed

# The digital divide

(Lind & Boomgarden, 2019; Tichenor, Donohue, & Olien, 1970)

- If the adoption of mass media in a social system increases, people with a high socio-economic status benefit more than people with lower status segments
- New technologies do not lead to an equalization of society but to deepened “knowledge gaps”
- Meta-analyses show a small to medium correlation between educational level and (factual) knowledge ( $r = .26$ )
- Small but constant moderator effects of media use: Educated people benefit more than uneducated people ( $r = .08$ )
- Mainly online media and newspapers, less so for television (but who uses linear TV today...)

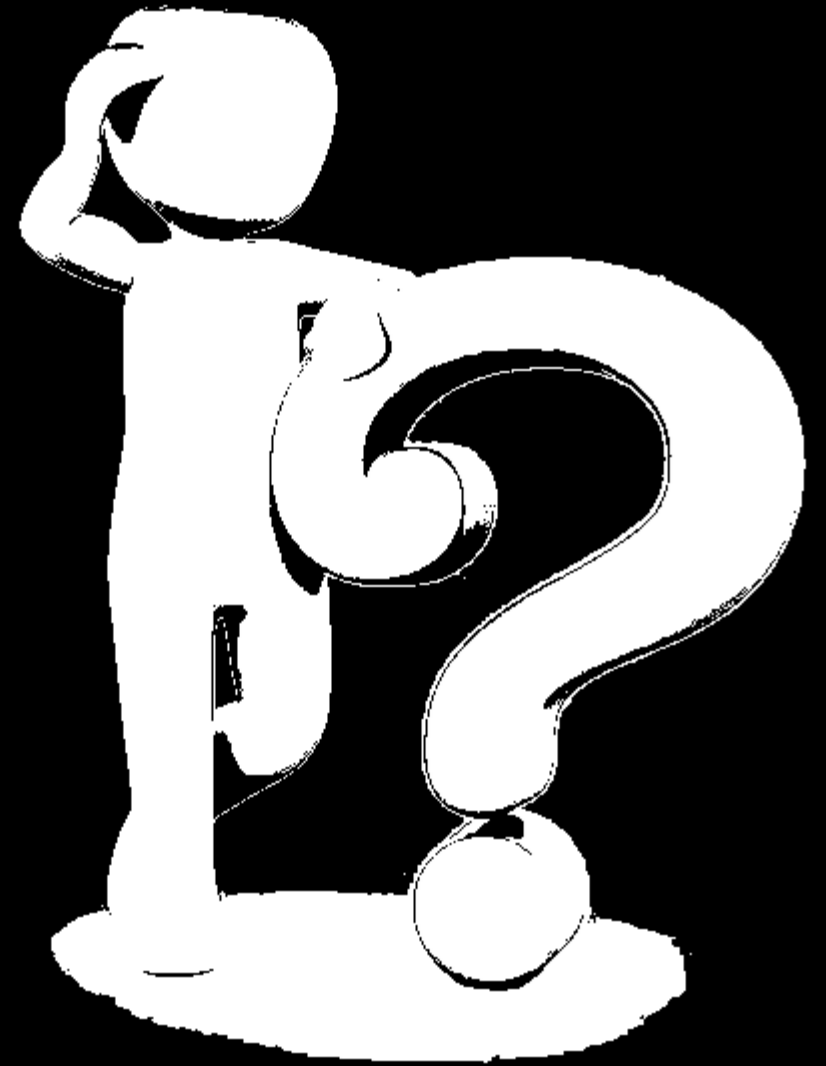


# The IOT leads to new digital divides

(van Deursen et al., 2019; 2021)

- Survey of the Dutch population ( $N = 1356$ )
- Almost half had an IoT device, especially smartwatches, thermostats and lighting systems
- People with low incomes/low education had a less positive attitude towards the IoT & fewer devices
- But even those who owned smart technologies often did not fully understand these devices
- Similar relationships also shape the access to technology more generally – plus, the majority population often has the most access to digital technologies

**Questions: Write me on  
itslearning**

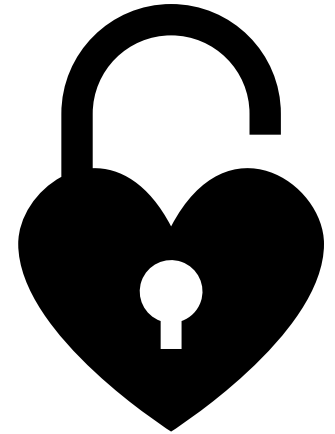


# **The Internet of Things and Privacy**

# Privacy

(General Data Protection Regulation, 2016; Trepte 2020)

- The ability to control access to information about one's life and personal space
- „The protection of natural persons in relation to the processing of personal data is a fundamental right. Article 8(1) of the Charter of Fundamental Rights of the European Union (the ‘Charter’) and Article 16(1) of the Treaty on the Functioning of the European Union (TFEU) provide that everyone has the right to the protection of personal data concerning him or her.“
- The processing of personal data should be designed to serve mankind. The right to the protection of personal data is not an absolute right; it must be considered in relation to its function in society and be balanced against other fundamental rights [...]



# Privacy is a key concern for the IOT

(Rose et al., 2015)

- **With regard to individual data streams**
  - Lack of awareness about data collection
  - Smart TV with voice control must constantly listen to recognize commands
- **With regard to the aggregation of data streams**
  - Fitness tracker says little movement,
  - Refrigerator only says fast food
  - Door lighting says door hasn't opened
  - Alexa reports that the voice is flat
  - => Depression?
- A study with 607 US-participants recently showed that 10 seconds of voice recordings were enough to detect Diabetes Type II with 75% AUC (for males, 71% for females)



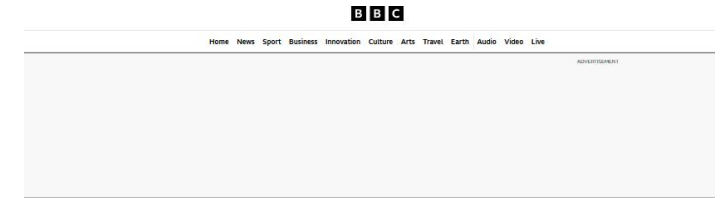
# Why is the IoT so special

(McCallum & Singleton, 2024; McShane, 2023; Rose et al., 2015)

- Informed consent often not explicitly requested
- IoT invades private spaces
- Often, multiple people are affected at the same time (e.g., sensors in public spaces)
- More data, more surveillance in private homes
- What could possibly go wrong?



Tony Scott, 1998



## Period trackers 'coercing' women into sharing risky information

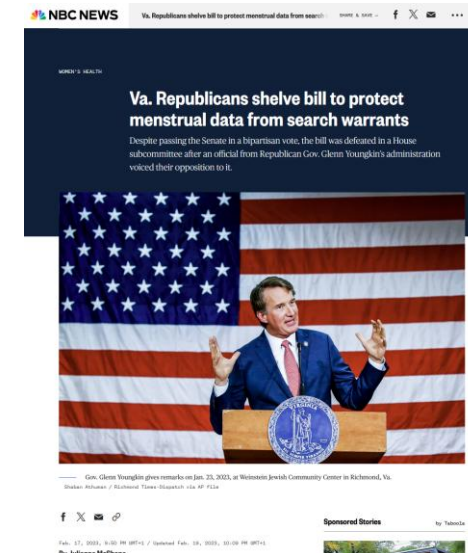
10 May 2024

Shiona McCallum & Tom Singleton  
Technology reporter

Share Save



Female health monitoring apps are putting women at risk by 'coercing' them into disclosing - and then poorly handling - highly sensitive data, according to new research.



# Why do people use the technology anyways?

(Trepte, 2020)

- The social media privacy model by Sabine Trepte
- Users compute an individual assessment of the
  - Levels of access to them in an interaction with others (others = persons, companies, institutions)
  - Availability of control mechanisms, whether it is interpersonal communication
  - Trust and norms regarding this specific type of interaction
- Self-disclosure is the result of this assessment



Sabine Trepte via Universität Hohenheim

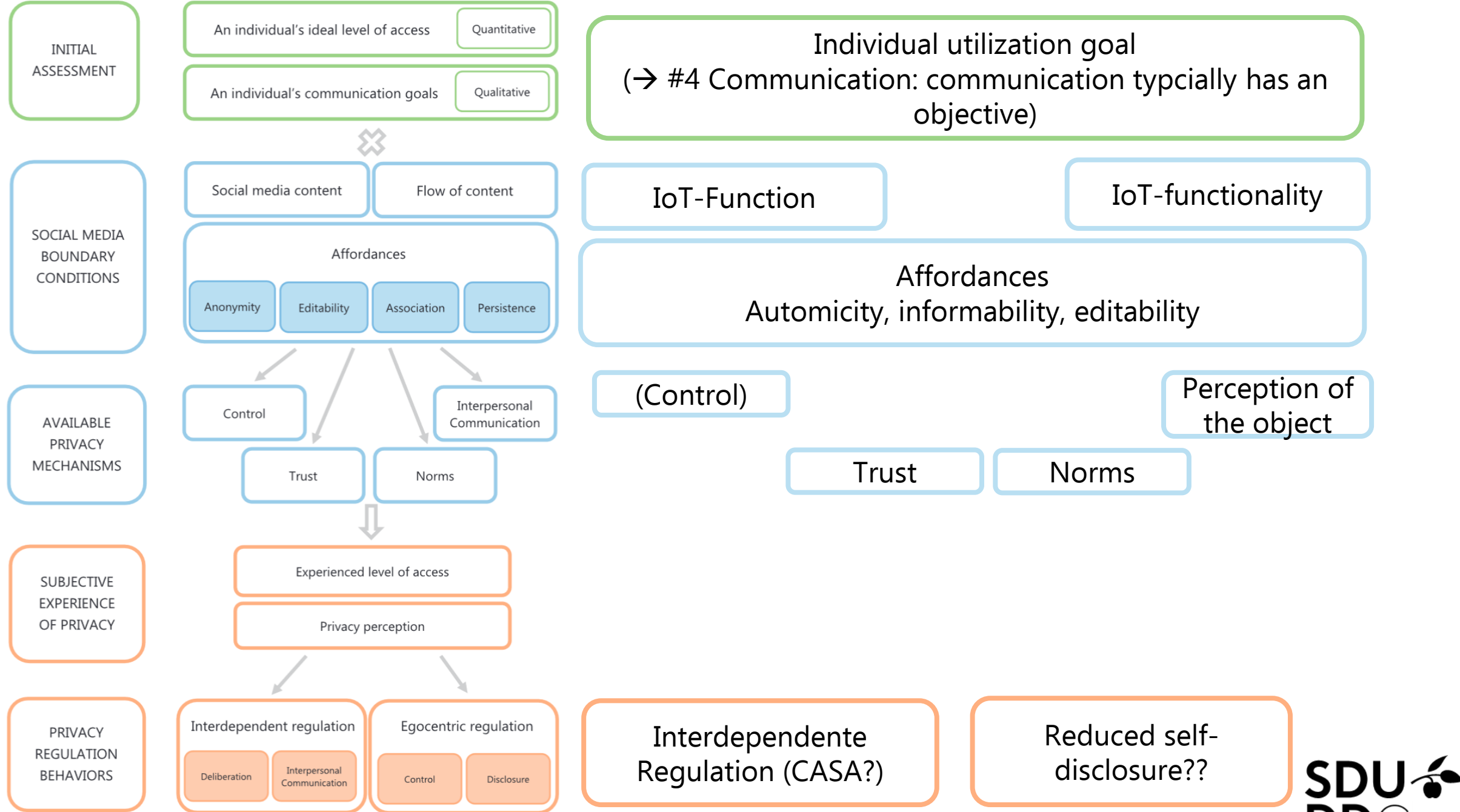


Figure 1 The social media privacy model.

# But not everyone can do it... (Livingstone et al., 2020; Mascheroni et al. 2020)

Children are increasingly monitored from parents, social media, smart toys, and other sources

- In 2017, "my friend Cyla" was banned and classified as a spying tool
- Parents feel under pressure between work, social obligations and norms of "good parenting" ("caring datavillance")

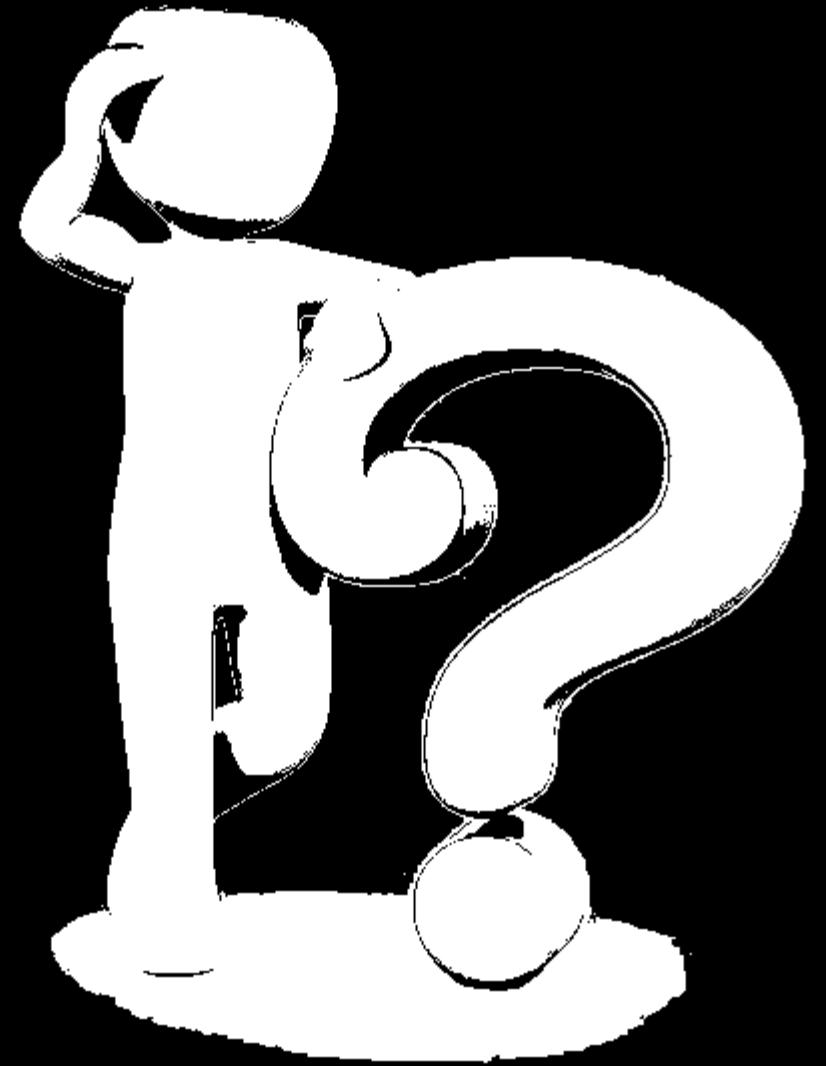
At the same time: Widespread sharenting – contrary to the right to privacy of children

Children are clever: they develop (within their means) attempts to circumvent surveillance – for example through fake accounts

But: Often children are often better at controlling interpersonal privacy than commercial privacy



**Questions: Write me on  
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# **AI 507: Artificial Intelligence and Society**



**#10 Humans in digital  
environments – a look at  
social media –  
guest talk by Anne Clausen**

# References

- Crawford, D. (2024, May 3). *What is AirTag stalking and how to protect against it*. Proton VPN. <https://protonvpn.com/blog/airtag-stalking>
- Elbéji, A., Pizzimenti, M., Aguayo, G., Fischer, A., Ayadi, H., Mauvais-Jarvis, F., Riveline, J.-P., Despotovic, V., & Fagherazzi, G. (2024). A voice-based algorithm can predict type 2 diabetes status in USA adults: Findings from the Colive Voice study. *PLOS Digital Health*, 3(12), e0000679. <https://doi.org/10.1371/journal.pdig.0000679>
- Hansen, A. R., Trotta, G., & Gram-Hanssen, K. (2024). Smart home technology adoption in Denmark: Diffusion, social differences, and energy consumption. *Energy Efficiency*, 17(3), 16. <https://doi.org/10.1007/s12053-024-10202-3>
- <https://eur-lex.europa.eu/eli/reg/2016/679/oj>
- <https://foodnationdenmark.com/cases/danish-farmers-lead-the-way-in-precision-agriculture/>
- <https://www.danskindustri.dk/brancher/di-digital/industri-4-0/>
- <https://www.youtube.com/watch?v=JHtEGCBgzjE>
- Intelligente Technologien finden sich in Wohnzimmern (79 Prozent), Gärten (51 Prozent) und Kinderzimmern (24 Prozent)
- Jha, S. N. (2022, December). Baby Monitor Market Analysis. *Fact.MR*. <https://www.factmr.com/report/626/baby-monitor-market>
- Kristensen, D. B., & Ruckenstein, M. (2018). Co-evolving with self-tracking technologies. *New Media & Society*, 20(10), 3624–3640. <https://doi.org/10.1177/1461444818755650>
- Kühl, E. (2017, Februar 17). My Friend Cayla: Vernichten Sie diese Puppe. *Die Zeit*. <https://www.zeit.de/digital/datenschutz/2017-02/my-friend-cayla-puppe-spion-bundesnetzagentur>
- Kumari, P., & Jain, A. K. (2023). A comprehensive study of DDoS attacks over IoT network and their countermeasures. *Computers & Security*, 127, 103096. <https://doi.org/10.1016/j.cose.2023.103096>
- Lind, F., & Boomgaarden, H. G. (2019). What we do and don't know: A meta-analysis of the knowledge gap hypothesis. *Annals of the International Communication Association*, 43(3), 210–224. <https://doi.org/10.1080/23808985.2019.1614475>
- Livingstone, S., Stoilova, M., Nandagiri, R., Milosevic, T., Zdrodowska, A., Mascheroni, G., Zaffaroni, L. G., Cino, D., & Wartella, E. A. (2020). The datafication of childhood: Examining children's and parents' data practices, children's right to privacy and parents' dilemmas. *AIIR Selected Papers of Internet Research*. <https://doi.org/10.5210/spir.v2020i0.11136>
- McCallum, S., & Singleton, T. (2024). Period trackers 'coercing' women into sharing risky information. *BBC*. <https://www.bbc.com/news/articles/cmj6j3d8xjjo>
- McShane, J. (2023, February 19). *Va. Republicans shelve bill to protect menstrual data from search warrants*. NBC News. <https://www.nbcnews.com/health/womens-health/va-republicans-shelve-bill-protect-menstrual-data-search-warrants-rcna71167>

# References

- Moore, A. (2022, September 5). 'I didn't want it anywhere near me': How the Apple AirTag became a gift to stalkers. *The Guardian*.  
<https://www.theguardian.com/technology/2022/sep/05/i-didnt-want-it-anywhere-near-me-how-the-apple-airtag-became-a-gift-to-stalkers>
- Paiva, L. D. S., Oliveira, J., & De França, T. C. (2023). The use of wearables to prevent sexual violence: A systematic mapping study. *Proceedings of the XIX Brazilian Symposium on Information Systems*, 260–267. <https://doi.org/10.1145/3592813.3592913>
- Paulsen, N. (2022). 43 Prozent der Deutschen nutzen Smart-Home-Technologien | Presseinformation | Bitkom e.V. *Bitkom*. <https://www.bitkom.org/Presse/Presseinformation/Smart-Home-2022>
- Rose, K., Eldridge, S., & Chapin, L. (2015). *The Internet of Things: An overview*. The Internet Society.
- Seidel, S., & Berente, N. (2020). Automate, informate, and generate: Affordance primitives of smart devices and the internet of things. In S. Nambisan, K. Lyytinen, & Y. Yoo (Eds.), *Handbook of Digital Innovation*. Edward Elgar Publishing. <https://www.e-elgar.com/shop/gbp/handbook-of-digital-innovation-9781788119979.html>
- Stentoft, J., Rajkumar, C., & Madsen, E. S. (2017). *Industry 4.0 in Danish industry: An empirical investigation of the degree of knowledge, perceived relevance and current practice*. University of Southern Denmark.
- Tichenor, P. J., Donohue, G. A., & Olien, C. N. (1970). Tichenor, P. J., Donohue, G. A., & Olien, C. N. (1970). Mass media flow and differential growth in knowledge. *The Public Opinion Quarterly*, 34(2), 159–170.
- Trepte, S. (2020). The social media privacy model: Privacy and communication in the light of social media affordances. *Communication Theory*, qtz035. <https://doi.org/10/ggwr6>
- Van Deursen, A. J. A. M., Van Der Zeeuw, A., De Boer, P., Jansen, G., & Van Rompay, T. (2021). Digital inequalities in the Internet of Things: Differences in attitudes, material access, skills, and usage. *Information, Communication & Society*, 24(2), 258–276. <https://doi.org/10.1080/1369118X.2019.1646777>
- Van Deursen, A. J., & Van Dijk, J. A. (2019). The first-level digital divide shifts from inequalities in physical access to inequalities in material access. *New Media & Society*, 21(2), 354–375. <https://doi.org/10.1177/1461444818797082>
- Van Deursen, A. J., & Van Dijk, J. A. (2019). The first-level digital divide shifts from inequalities in physical access to inequalities in material access. *New Media & Society*, 21(2), 354–375. <https://doi.org/10.1177/1461444818797082>